# HARIHARASUTHAN S

AI / ML Engineer

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LinkedIn • GitHub • Portfolio

## PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with 2+ years of production experience in Generative AI, RAG systems, Computer Vision, and LLM Fine-tuning. Delivered measurable outcomes including a 40% reduction in CAD re-texturing time and a 50% drop in manual inspection time for industrial defect detection. Proficient in Python, PyTorch, TensorFlow, FastAPI, and LangChain.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, C, SQL |
| **ML / DL** | PyTorch, TensorFlow, Scikit-Learn, Unsloth |
| **Generative AI** | GANs (Pix2Pix, CycleGAN), LLM Fine-tuning, RAG, Prompt Engineering |
| **NLP & Speech** | LangChain, SpaCy, Hugging Face Transformers, PyAnnote (Diarization) |
| **Computer Vision** | OpenCV, Image Segmentation, Object Detection, Defect Detection |
| **MLOps & APIs** | FastAPI, Pydantic, Git/GitHub, REST APIs, Docker |
| **Data & Viz** | NumPy, Pandas, Matplotlib, Seaborn, Plotly |
| **Cloud & Tools** | Microsoft Azure (AZ-900), Google Cloud, VS Code, Jupyter, Colab |

## PROFESSIONAL EXPERIENCE

### Adela Software and Services Pvt. Ltd. | AI / ML Engineer

*Sep 2025 – Present • Trichy, India*

- Developed production-grade RAG pipelines using LangChain for enterprise document Q&A.;
- Built FastAPI-based ML inference services with async handling and Pydantic schema validation.
- Improved API reliability by 30% through error handling and environment-based configuration.
- Achieved >90% accuracy on internal classification, regression, and NLP benchmarks.
- Reduced new engineer onboarding time by 25% via clean architecture and CI-ready patterns.

***Tech Stack:*** *Python, PyTorch, TensorFlow, LangChain, FastAPI, RAG, ChromaDB*

### Nilatech Robotics and Automation Pvt. Ltd. | AI / ML Engineer

*Sep 2024 – Aug 2025 • India*

- Deployed Pix2Pix GAN pipeline for CAD re-texturing, reducing processing time by 40%.
- Built CNN + OpenCV defect detection system for brake components, cutting manual inspection time by 50%.
- Engineered end-to-end ML lifecycle — data ingestion, preprocessing, training, evaluation, deployment.
- Integrated AI inference into robotic automation systems in collaboration with hardware engineers.

***Tech Stack:*** *Python, PyTorch, TensorFlow, GANs, OpenCV, Computer Vision*

## KEY PROJECTS

### RAG Knowledge Assistant — Adela Software and Services

- Built a Retrieval-Augmented Generation system using LangChain, ChromaDB, and Sentence Transformers for enterprise document Q&A.;

- Implemented hybrid retrieval with reranking, achieving sub-2s response latency at scale.
- Reduced hallucinations through optimized context retrieval and chunking strategies.
- Deployed as a FastAPI microservice with async endpoints and Pydantic request validation.

  ***Tech Stack:*** *Python, LangChain, ChromaDB, Sentence Transformers, FastAPI, PyTorch*

### CAD Re-Texturing using Pix2Pix GAN — Nilatech Robotics

- Designed and trained a Pix2Pix GAN pipeline to automate texture generation for manufacturing CAD models.
- Delivered a 40% reduction in re-texturing processing time and enabled real-time texture previews for clients.
- Managed full training pipeline including paired dataset curation, augmentation, and FID score evaluation.

  ***Tech Stack:*** *Python, PyTorch, Pix2Pix GAN, OpenCV, TensorFlow*

### Industrial Defect Detection System — Nilatech Robotics

- Built a CNN-based visual defect detection system for brake components using OpenCV preprocessing.
- Reduced manual inspection time by 50% and improved defect catch rate with high-precision classification.
- Engineered reproducible training pipeline with automated data ingestion, augmentation, and validation splits.

  ***Tech Stack:*** *Python, PyTorch, CNN, OpenCV, TensorFlow, Scikit-Learn*

### AI-Powered Audio Diarization System — Personal Project | GitHub

- Built a real-time speaker diarization system using PyAnnote 3.1 for accurate multi-speaker segmentation.
- Fine-tuned segmentation model improving speaker boundary detection accuracy by ~18%, reducing DER on benchmarks.
- Applicable to meeting transcription, legal proceedings, and call-centre analytics; reduces manual transcription cost by up to 70%.

  ***Tech Stack:*** *Python, PyAnnote 3.1, Hugging Face Transformers, FastAPI*

## EDUCATION

### K. Ramakrishnan College of Engineering

*B.Tech, Computer Science and Business Systems • Sep 2021 – May 2025 • First Class Honours*

## CERTIFICATIONS

- IBM – Generative Artificial Intelligence
- Microsoft – Azure Fundamentals (AZ-900) — verify.certiport.com: wkXY9-2FqS
- Coursera – Machine Learning for All
- Google – Google Cloud Essentials

## ACHIEVEMENTS & RECOGNITION

- 1st Runner-Up — National Level Hackathon (AI/ML Track)
- Finalist — Accenture Innovation Hackathon (International Level)
- Participant — PEC Hackathon (International Level)
- Patent Application filed for AI-based Disaster Headcount & Public Safety System (Neomat)